

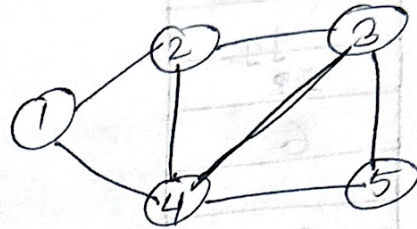
GRAPHS

Non Linear Data Structure

① Representation $\rightarrow$

① Adjacency Matrix

② " List



① Adjacency Matrix $\Rightarrow n \times n$

Adj

	1	2	3	4	5
1	0	1	0	1	0
2	1	0	1	1	0
3	0	1	0	1	1
4	1	1	1	0	1
5	0	0	1	1	0

It is a Matrix $A[n][n]$ where n is no. of vertices

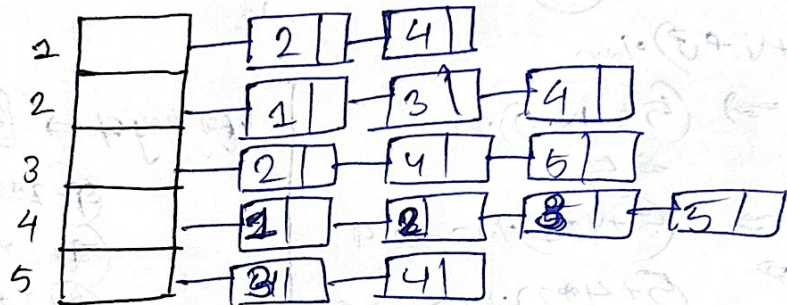
$A[i][j] = 1$ if i & j are adjacent
 $= 0$ otherwise

$\Rightarrow O(n^2)$

② Adjacency List $\Rightarrow$

~~$O(n^2)$~~

$O(n + 2e)$
edge

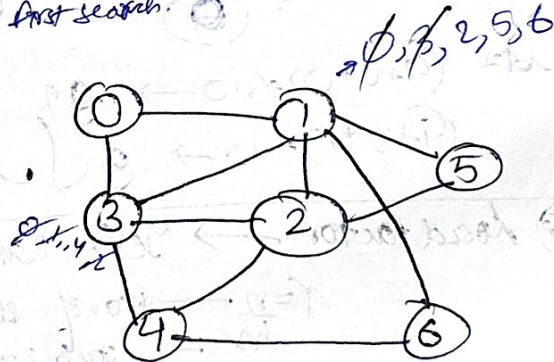
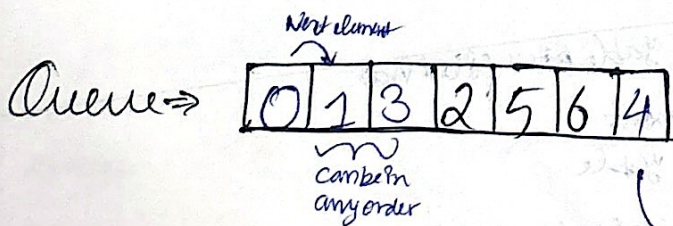


③ Graph Traversals $\rightarrow$

① BFS $\Rightarrow$ Breadth First Search

② DFS $\Rightarrow$ Depth First Search

① BFS $\rightarrow$ Queue Data Str is used



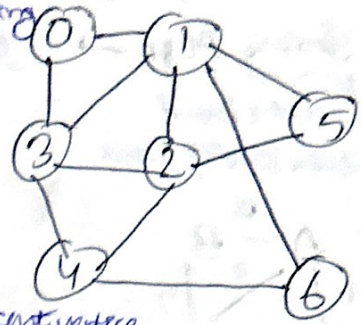
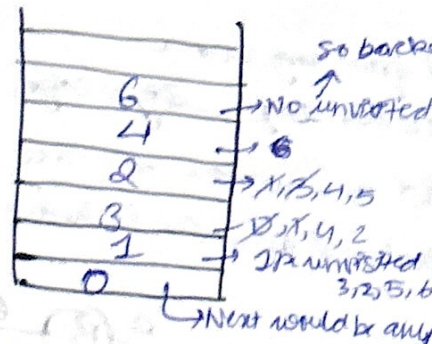
For 1 $\rightarrow$ 0, 3, 2, 5, 6

See adjacent unvisited elements
and put in queue

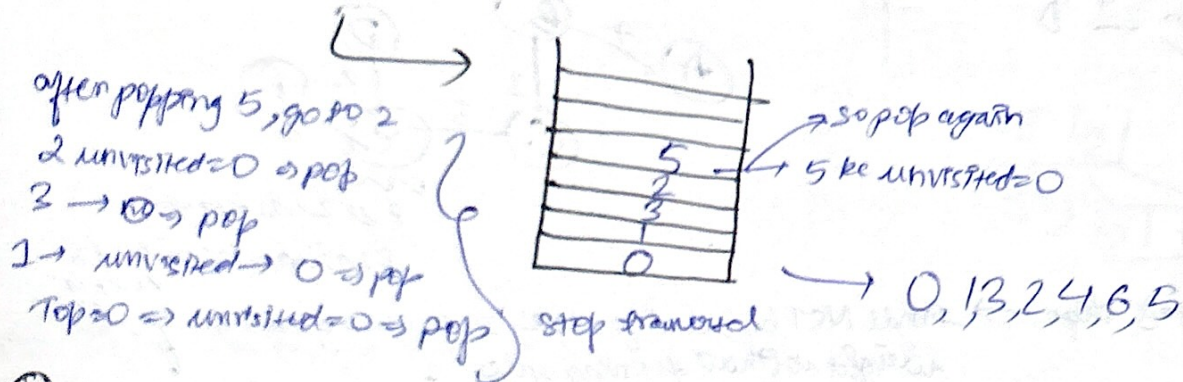
Result $\Rightarrow$ 0 1 3 2 5 6 4

② DFS $\rightarrow$ Stack Data Structure

Stack $\Rightarrow$



Backtracking $\rightarrow$ Pop last inserted element, here 6
 4 has no vertex unvisited so pop out
 2 has 5 has unvisited



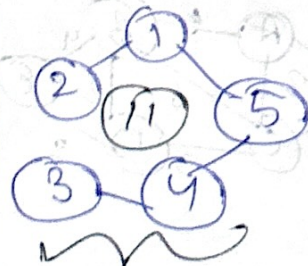
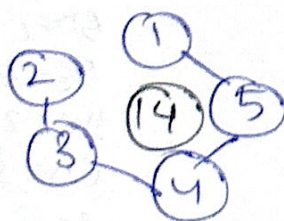
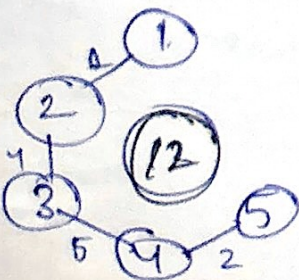
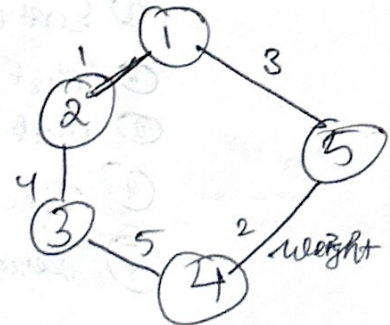
③ Minimum Spanning Tree $\rightarrow$

Graph = $G(V, E)$

Spanning Tree = $G'(V', E')$

Vertex = $V' = V \rightarrow E' \subset E$
 edge of ST is subset of graph

$$E' = |V| - 1$$



min of them all

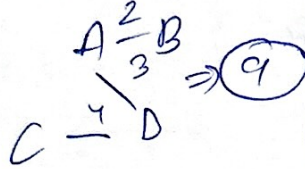
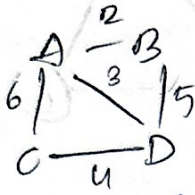
A Min Spanning Tree is a spanning tree of a weighted connected graph having min total edge weight.

① contains $n-1$ edges,

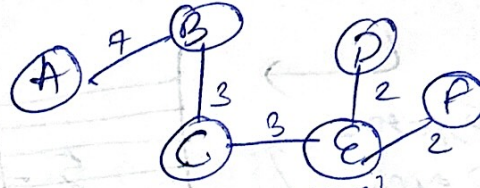
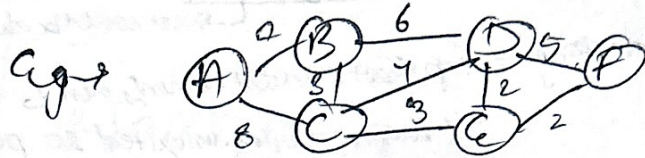
② out of all possible choose one with min total weight

- ② Vertices must be connected
- ③ MST is always acyclic

④ Prim's Algo →
Vertex based
Uses Priority Queue



Start from any vertex
Select min edge nearest edge
Add new vertex to Tree
Repeat until all vertices included

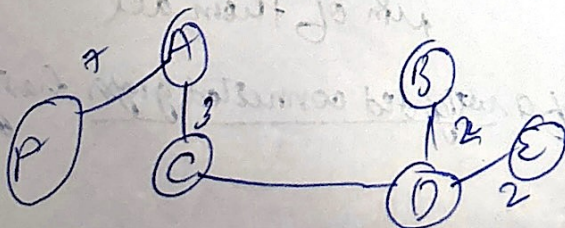
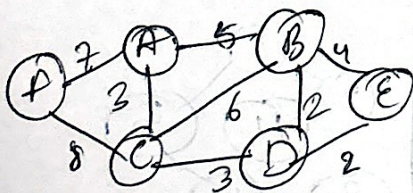


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⑤ Kruskal's Algo → Find MST by selecting edges in increasing order of weight without forming cycles.

- ① Sort all edges
- ② Select element edge
- ③ Check cycle
- ④ If no cycle → include
- ⑤ Repeat until $n-1$ edge



- $BD=2$ ✓
- $DE=2$ ✓
- $AC=3$ ✓
- $CD=3$ ✓
- $\alpha BE=4$
- $\alpha AD=5$
- $\alpha BC=6$
- $\alpha AE=7$ ✓
- $\alpha PC=8$

① B Tree $\longrightarrow$

② Order 3 by inserting values from 1 to 10.

$m=3$, Max key $= 3-1 = 2$

